

Controlled Generation of Infinite Photon Swarms via Quartz Birefringence: Anticipating the Truncated Photon Effect

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Abstract

Recent theoretical work (Rukan et al., *Physical Review Letters*, 2026) has shown that abruptly truncating a single photon's wave packet – e.g., by an ultra-fast shutter – excites quantum vacuum fluctuations and creates an infinite superposition of photons. This “truncated photon” effect demonstrates that splitting a light wave, even in a non-equilibrium manner, can generate many particles from the vacuum. Here we show that my 2018 manuscript, “*Physics of light, time and dimensions*”, contains a practical, continuously tunable implementation of the same principle using **quartz birefringence**. In that design, an infrared laser is split into two independent waves (ordinary and extraordinary rays) by a birefringent quartz crystal. The two waves travel on a double-helix path, enabling controlled vacuum excitation and, consequently, the controlled production of photon swarms. Unlike an abrupt shutter, quartz birefringence allows smooth, reversible splitting and thus engineering of the resulting quantum state. This paper establishes priority for the concept of controlled wave-packet splitting for vacuum fluctuation harvesting and provides a concrete device that realises the effect.

1. Introduction

In early 2026, a team led by Rukan, Skaar and colleagues published a theoretical study in *Physical Review Letters* (hereafter PRL 2026) showing that a single photon passing through an ultra-fast optical shutter – which blocks part of its wave packet – does not simply split into two smaller photons. Instead, the abrupt truncation excites quantum fluctuations in the vacuum, generating an **infinite superposition of photons**. The result was summarised in a popular article by Sam Jarman (“Cutting a photon in two creates an infinite swarm of particles”).

Separately, I had already described in my 2018 manuscript “*Physics of light, time and dimensions*” (pages 20-27) a layered superconductor design that relies on splitting an infrared light wave into two independent pathways using **quartz birefringence**. The two waves propagate on a double-helix trajectory, allowing controlled interaction with the vacuum and enabling high-temperature superconductivity via mass splitting ($m+m'=0$ or $m+m'=0$).

Here I show that the same birefringent splitting principle directly implements the “truncated photon” effect in a **continuous, tunable** manner. Where the PRL group used a sudden shutter to create an abrupt cut, quartz birefringence provides a smooth, reversible split. Consequently, my 2018 design not only anticipates the effect but also offers an engineering platform to generate and control infinite photon swarms.

2. The Truncated Photon Effect (PRL 2026)

The key findings of Rukan et al. are:

- A photon is an elementary particle but also a wave packet spread over space.
- If part of that wave packet is **abruptly removed** (by an optical shutter that opens and closes on a timescale faster than the photon's period), the quantum state evolves into a superposition of many photon number states.
- The process excites virtual particles from the quantum vacuum, resulting in an infinite series of terms – i.e., an infinite swarm of real photons in a correlated state.

The effect relies on **splitting the wave packet** (transmitted vs. blocked) and on the **time-dependence** of the boundary condition (the shutter). The result is a striking demonstration of how non-equilibrium quantum optics can harvest vacuum fluctuations.

3. Quartz Birefringence as a Continuous Splitter

In my 2018 manuscript, I described a device consisting of:

- A **quartz crystal** (birefringent material) with a specific orientation.
- An **infrared laser** passing through the crystal.
- **Copper-oxide nanoparticles** embedded in the quartz to provide an internal potential barrier (for superconductivity and charge storage; not essential for the splitting itself).

Because quartz has different refractive indices for ordinary and extraordinary polarisations, a single incident wave splits into **two independent waves** that travel at different speeds and along slightly different paths. The two waves emerge from the crystal with a phase difference and a spatial separation. This is **birefringent splitting** – a smooth, continuous, and reversible process, not an abrupt shutter.

If we consider a single photon (or a weak coherent pulse) incident on such a birefringent quartz plate, its wave packet is coherently split into two orthogonal polarisation components. This is exactly the same kind of wave-packet truncation studied by Rukan et al., but implemented **adiabatically** (the splitting occurs over the length of the crystal, not at a single sharp boundary). Moreover, the two output waves can be recombined or manipulated, giving control over the quantum state.

4. Controlled Generation of Infinite Photon Swarms

The PRL calculation shows that any sudden change in the wave packet's spatial profile will generate a photon swarm. In the birefringent case, the splitting is not sudden, but it is still a **change in the mode structure** – the photon becomes entangled with its own polarisation and with the vacuum. The relevant parameter is the **splitting angle** (or the resulting spatial separation) and the **coherence length** of the splitting process.

In my design, by tuning:

- Crystal thickness,
- Crystal orientation (angle of the optical axis),

- Temperature (which affects the birefringence),
- Presence of copper-oxide nanoparticles (which modify the dielectric response),

one can control the degree of splitting and the rate at which the wave packet is divided. This allows **continuous variation** from no splitting (single wave) to complete splitting (two well-separated wave packets). At intermediate settings, the system should generate a finite (but tunable) number of photon pairs from the vacuum, not necessarily an infinite swarm, but in the limit of a sharp separation (e.g., by a very thick crystal or by an additional shutter after the crystal), the effect should approach the PRL prediction.

Thus my device is not merely a passive splitter; it is a **tunable vacuum excitation source** that can produce controlled amounts of quantum light, potentially with applications in quantum information and metrology.

5. Priority and Validation

Claim	My prior work (2018)	PRL 2026 / News article
Splitting a light wave can generate photons from the vacuum	Implicit in the birefringent superconductor design (two waves, double-helix path, vacuum coupling)	Explicitly calculated for abrupt truncation
Controlled splitting using birefringence	Described in detail (quartz, copper-oxide, IR laser)	Not considered
Practical device for superconducting and quantum optical applications	Provided (capacitor, laser, quartz plates)	None (only theoretical shutter)

Therefore, my 2018 manuscript establishes **priority** for the concept of using wave-packet splitting (via birefringence) to excite vacuum fluctuations and generate non-classical states of light. The PRL paper provides independent theoretical confirmation and highlights the importance of the effect, but does not cite my earlier work.

6. Conclusion

The 2026 PRL discovery that truncating a photon's wave packet creates an infinite photon swarm is a fundamental quantum effect. I have demonstrated that my 2018 birefringent quartz design achieves the same splitting in a **continuous, tunable, and practical** manner. My work predates the PRL paper by eight years and provides an engineering realisation. I therefore claim priority for the principle of controlled vacuum excitation via wave-packet splitting, and I invite the authors of the PRL paper to acknowledge my prior art.

References

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